

The Container Program: A Research Summary

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*Internal program summary for N. Ghani (PI), Kodamai “AI Mathematician” grant. Every mathematical claim carries its honest registry trust-grade (*speculative* < *computed* < *proved* < *peer-reviewed* < *published* < *lean-verified*). No MacBeth result is self-published; the strongest social grade on our own work is *peer-reviewed* (by Rick). Grades track the registry node, not the surrounding prose. (The projects tree is not under version control, so no commit hash is quoted.)*

1 Thesis

The claim this program defends is that the four-way equivalence

$$\text{Containers} \simeq \text{Directed Containers} \simeq \text{Polynomial Comonoids} \simeq \text{Small Categories}$$

is the correct *foundation* for formally-verified composition of heterogeneous computational systems. A directed container is simultaneously a data shape, a comonad, a comonoid in the composition-monoidal structure $(\text{Cont}, \triangleleft, I)$, and a category; a system built from one is therefore a system whose dynamics, data layout, and type discipline are a single object that different tools can read differently without translation loss. Composition of such systems is then not an engineering convenience but an algebraic question with a yes/no answer.

The central technical tool for *when* two such systems compose is the **Zappa–Szép product** (equivalently, a distributive/matched-pair law): given directed containers C, D , a global product $C \bowtie D$ exists iff two conditions hold—a freeness condition (L) and a global-closure condition (G)—and the obstruction to (G) is a second cohomology class. When that class is nonzero, the composite does not exist on the nose, and the failure is observable: in blockchain re-entrancy it is exactly the generator of $H^2 \cong \mathbb{F}_2$. This is the program’s bridge from category theory to economically-consequential correctness.

The work is organised against the three grant pillars. **Theory** (§3): the equivalence-chain core, the tangent/Cartan arc for the container derivative, the Zappa–Szép / H^2 obstruction calculus, and the fullness/admissibility/change-of-base lane. **Applications** (§4): agent orchestration, blockchain re-entrancy, supply chains, linear attention, and game comonads, each realised as a directed container and each with its composition question answered. **Formalisation** (§5): roughly 107 sorry-free Lean 4 declarations, including the 1-categorical core of the equivalence chain and the arithmetic oracles of the tangent arc.

One honesty statement up front, because it governs the grant narrative. The equivalence chain is **lean-verified only at its 1-categorical core** (milestones M1–M4). The enriched rung (M5) is unclassified/open and the ∞ -rung (M6) is speculative, so the *full* chain is honestly *in-progress*. The pieces that are done are done to machine-checked standard; the chain is not yet closed end-to-end.

2 The spine

Four results carry the program. Stated crisply, with grades.

Result 1 (Equivalence-chain core, M1–M4). (lean-verified) The container representation functor is fully faithful (M1); comonad laws $\Leftrightarrow$ directed-container laws with both round-trips (M2); directed containers = comonoids in $(\mathbf{Cont}, \triangleleft, I)$ in both directions (M3); and $\mathbf{DCont} = \mathbf{Cof}$ as an isomorphism of categories at the morphism level (M4). All four are sorry-free Lean, axiom footprints in $\{\mathbf{propext}, \mathbf{Quot.sound}\}$. The *full* three-way equivalence root is **in-progress**: M5 (enriched $\mathbf{DCont} = \mathbf{Cof}$) is unclassified and M6 (an ∞ -level statement, Kan–Chen Conj. 7.2) is speculative.

Result 2 (Flagship T1: fullness $\Leftrightarrow$ unit-connectedness). (proved) The container extension $\llbracket - \rrbracket : \mathbf{Fam}(C^{\text{op}}) \rightarrow [C, C]$ is fully faithful **iff the monoidal unit I is connected** (i.e. $C(I, -)$ preserves coproducts). This corrects the natural “extensivity” conjecture: $\mathbf{Set} \times \mathbf{Set}$ is lexextensive yet $\llbracket - \rrbracket$ is not full, while $\mathbf{Set}$ is classical precisely because 1 is connected.

Result 3 (Tangent/Cartan arc for ∂). (proved; lean-verified arithmetic cores) The AAGM one-hole container derivative ∂ is the differential combinator of a Cartesian differential category / first-order tangent structure on $\mathbf{SPoly}$, realised by dual-number substitution $T(p) = p(y + \varepsilon v) \bmod \varepsilon^2$ (proved; 0-axiom finite Lean bridge). Among representable first-order tangent structures there are **exactly two**—the trivial one and the dual-number one (proved). And ∂ sits on a sharp **N-vs- $\mathbb{Z}$ Cartan knife**: over the rig $\mathbb{N}$ there is *no* scalar ring object and no fiberwise negation, so the full Cartan calculus does not descend (proved; $[\mathbf{propext}, \mathbf{Quot.sound}]$ core); group-completing $\mathbb{N} \rightarrow \mathbb{Z}$ switches it *on*— $\mathbf{SPoly}_{\mathbb{Z}}$ is Rosický, carries a scalar ring object, and meets the full Aintablian–Blohmann specification (proved).

Result 4 (Pairwise Zappa–Szép criterion and the (G) obstruction). (proved; lean-verified instantiation) A global Zappa–Szép product exists iff (L) freeness and (G) global closure both hold (proved); the (G)-obstruction is a class $[\omega] \in H^2$ (proved). Instantiated in agent orchestration / blockchain re-entrancy: unprotected re-entrancy is exactly (G)-failure with $[\omega(K_{\varepsilon})] = \varepsilon$ the generator of $H^2 \cong \mathbb{Z}/2$ (lean-verified).

3 Theory

3.1 The equivalence-chain core (Lane A)

Beyond Result 1, the chain’s infrastructure is largely machine-checked: matched-pair associativity $\Leftrightarrow$ the four Zappa–Szép axioms ZS1–ZS4 (lean-verified); monoidal coherence (pentagon + triangle) for all four tensors $\otimes, \times, +, \triangleleft$ on $\mathbf{Cont}$ (lean-verified); and the pairwise ZS criterion $(L) \wedge (G)$ (proved), shared into the chain.

Supporting structural results. The free monad on a container is the free $\triangleleft$ -monoid under grafting, with the three monoid laws machine-checked (lean-verified) (the universal property is *partial*: a multiplication-backward node and backward-uniqueness remain). The cofree comonad is the $\triangleleft$ -comonoid of p -trees with its couniversal property in container coordinates (proved). The state object ΔS is the codiscrete category, and $\mathbf{Workers}$ form a $(\mathbf{Set}, \times)$ -graded category via the Dirichlet tensor (proved).

Remark 5 (Honest status of the chain). The 1-categorical core is lean-verified; the enriched (M5) and ∞ (M6) rungs are open. The grant narrative should read “1-categorical core lean-verified, enriched/ ∞ open,” not “equivalence chain proved.”

3.2 The tangent / Cartan arc (Lane B)

The arc develops Result 3 into a full dividing-line picture.

Existence. $T = (-) \otimes W$, base change along $k \rightarrow \mathbb{D} = k[\varepsilon]/\varepsilon^2$ (Weil algebra $W = \mathbb{N}[\varepsilon]/\varepsilon^2$), is a strict $\triangleleft$ -monoidal functor whose laxator is the AAGM chain rule, and it restricts to $\mathbf{DCont}$ (proved). Honest caveat: the target base is genuinely $\mathbb{D}$; over $\mathbf{Set}$ the tangent bundle carries no comonad structure.

Uniqueness. Among representable first-order tangent structures there are exactly two (trivial $W = \mathbb{N}$; dual-number $W = \mathbb{N}[\varepsilon]/\varepsilon^2$) (proved). **Rick referee correction (accepted):** the ambient category is base change by the Weil algebra, $T = (-) \otimes W$, *not* $T = (-) \triangleleft \mathbb{D}$ on full $(\mathbf{Poly}, \triangleleft)$. The $\triangleleft$ -reading was refereed wrong (on full $\triangleleft$ - $\mathbf{Poly}$ the honest operation collapses to $\{\text{id}\}$; $\triangleleft = \otimes W$ only on the linear layer). The write-ups were relabelled accordingly, and the former “third structure at infinite support” (F4) is **demoted to open**. Several Lean oracles support the uniqueness arithmetic: solid-idempotent collapse, universality rank-rigidity, and the two-model distinction, all [propext].

The $\mathbb{N}$ -vs- $\mathbb{Z}$ Cartan knife. Over the rig $\mathbb{N}$, $\mathbf{SPoly}/\partial$ has no scalar ring object and is not Rosický (no fiberwise negation), so the full Cartan calculus does *not* descend—only a rig-scalar fragment survives (proved; [propext, Quot.sound] arithmetic core). Group-completing to $\mathbb{Z}$ puts us on the far side of the knife: $\mathbf{SPoly}_{\mathbb{Z}}$ is Rosický, carries scalar ring object $R = \mathbb{Z}$, and satisfies the entire Aintablian–Blohmann spec, with Cartan descent a **computed** downstream consequence (proved; lean-verified positive companion).

A dividing line by refutation. The Lanfranchi “parallelizable $\Leftrightarrow$ Lie group object” port is **ill-posed** on polynomial functors (T is not Cartesian there); what survives—tangent-regular $\Leftrightarrow$ out-degree homogeneous—is *blind to composition*, hence not a groupoid criterion. Refuted both ways (proved, as an impossibility result).

3.3 Zappa–Szép / skew-brace / H^2 obstruction (Lane C)

The obstruction calculus behind Result 4. The solvability obstruction $o_\nu: H_+^2 \rightarrow C^3/A(Z_\circ^2)$ is a well-defined homomorphism killing B_+^2 with $\text{im } \varphi^+ = \ker o_\nu$: an additive class extends to a skew-brace cocycle iff $o_\nu = 0$ (proved; lean-verified at $|G| = 8$ slice and $|G| = 16$ lock). This is the honest repair of the earlier (refuted) $[\nu] \perp [\Omega]$ independence claim—the residue, σ , and $[\beta]$ are coupled.

The holonomy-composition ZS bridge is **peer-reviewed**: composite liftings compose by ZS product, isotropy containment holds, and the aligned-abelian $[\omega]$ obstruction is proved (the originally-stated part (b) was refuted and replaced by (b')). The two $[\omega]$ -sites (handoff vs. stabiliser) are irreducibly distinct (proved); emergent-holonomy meeting count = $\dim \text{Ext}^0$ with the higher Ext tower vanishing (proved). The cyclic H^2 engines— $H^2(C_n; \mathbb{Z}) = \mathbb{Z}/n$, $H^2(C_n; \mathbb{Z}/m) = \mathbb{Z}/\gcd(n, m)$, $H^2(\mathbb{Z}/2; \mathbb{Z}/2) = \mathbb{Z}/2$ with split/nonsplit witnesses—are (lean-verified) (Bézout built from scratch). Two roots sit honestly at **computed**: the groupoid ZS obstruction (the strong conjecture “groupoid $\Rightarrow [\omega] = 0$ ” was *refuted*; the correct dividing line is cohomological dimension ≤ 1) and the quiver skew-brace / Ferri-prunability separation (prunability = descent/subfunctor, Schreier-blind; many proved sub-nodes, including the peer-reviewed $[\Omega] = \text{Rathee–Yadav identification}$, under a **computed** root).

3.4 Fullness / admissibility / closed structure (Lane D)

Flagship T1 (Result 2) anchors this lane. Around it: the copower test yields the writer monad and, over any extensive base, decides fullness (proved) (residual Q' open); $(-) \triangleleft q$ has a left adjoint iff the unit is connected—T1’s condition applied twice—and over $\mathbf{Vec}$ the left adjoint exists iff verticality

(proved) (this root carries the large Gap-S / Conj. 6.2 ZFC sub-program); fibredness of $(-) \triangleleft q$ is *not* its left-closure (a two-sided separation, answering BHM) (proved).

Closed-structure classification: Dirichlet $\otimes$ on $\mathbf{Fam}(C^{\text{op}})$ is closed iff Φ is familially representable (T2) (proved); $\triangleleft$ -left-closure exists over a tiny/rigid base via the collapse $\triangleleft = \otimes$ (T4-left) (proved); a Day-convolutional tensor is left-closed iff $(-) \star B$ is polynomial (proved); $(-) \triangleleft q$ is a parametric right adjoint for every q (proved). Logic of containers: $\text{Cont}(\text{cod}) = \mathbf{Fam}(\text{cod}^{\text{op}})$ is a bifibration, quantifiers are Neil’s A/E liftings, and it is a co-hyperdoctrine with one-sided Beck–Chevalley/Frobenius (proved). Comonoid classifications: bare $\otimes$ -comonoids in $\mathbf{Poly}$ are families of monoids (answering Niu–Spivak Ch. 9 Q5), $\otimes$ -monoids are monoid + oplax functor, and double comonoids in $(\mathbf{Poly}, \triangleleft, \otimes)$ are sets of commutative monoids (Eckmann–Hilton) (proved; mostly lean-verified). The full Hedges 4×4 interchange table is computed with two corrections to Hedges (proved; most cells lean-verified). Two roots remain open/computed: the closed-convolutional classification ($\star \cong \times$ or $\vee_S$; bounded-arity proved, infinite-arity open) is **in-progress**, and the $\ltimes/\rtimes$ Dialectica tensors (with $\ltimes$ left-closed and the pair duoidal/LDC, both proved) sit under a **computed** root.

3.5 Containers over $\mathbf{Vec}$ / M-containers / change-of-base (Lane E)

Linear containers $\mathbf{Fam}(\mathbf{Vec}^{\text{op}})$: biproduct collapse $\llbracket S, P \rrbracket \cong \text{Id}^N$, $\llbracket - \rrbracket$ is *not* full over $\mathbf{Vec}$ (the extensivity crux again), and $\triangleleft$ -comonoids are families of k -algebras (proved; several lean-verified). M-containers: $M\text{-Cont} = \mathbf{Fam}(\mathbf{Kl}(M)^{\text{op}})$, with $\llbracket - \rrbracket_M$ always faithful and full iff M is codense and each $(M-)^A$ is connected (the coproduct criterion was refuted; probabilistic D keeps fullness) (proved; THM3 composition-polynomiality sub-tree carries many Rick-peer-reviewed nodes). Change-of-base: $\mathcal{CL} \cap \mathcal{UN} = \{\text{Id}\}$ —no nontrivial effect monad gives a uniform fully-faithful enrichment theorem, a sharp negative answer to Neil UID-168 (peer-reviewed; lean-verified algebraic core). Finally the monad→comonad transfer $G(S, P) = (S, M \circ P)$ (comonad laws $\Leftrightarrow$ monad laws, = left coclosure/Lan) (proved); a mixed distributive law (entwining) exists for all M with the reverse failing on branching (proved); and effect-coeffect arrows form a category iff M is non-branching, with λ a Turi–Plotkin bialgebra for all M (proved; the largest tree, ~ 90 nodes, many lean-verified).

4 Applications

- **Agent orchestration = Zappa–Szép weld.** An orchestration is a directed container; two orchestrations compose as $C \bowtie D$ iff $\varepsilon = 0$. Unprotected re-entrancy is (G)-failure (proved; peer-reviewed standalone note).
- **Blockchain re-entrancy = failed distributive law.** The re-entrancy obstruction is $[\omega(K_\varepsilon)] = \varepsilon$, the generator of $H^2 \cong \mathbb{Z}/2$ (lean-verified). This is the program’s sharpest Applications claim: a known class of economically-consequential smart-contract bugs is exactly a nonzero second cohomology class. Security audits of such contracts are, in this framing, searches for failed distributive laws.
- **Supply chains = directed containers.** A supply chain is a directed container; a ZS product governs composition; inventory inconsistency is $[\omega] \in H^2(\mathbf{Sk}; \mathbb{Z}/n)$, with a worked ε -family computed and $n = 2$ reusing the orchestration bit (computed).
- **Linear attention and GA composition.** Linear attention is the $\odot$ product in $\mathbf{Mat}(\mathbf{Vec})$ *exactly at fixed context*; the strong “stack = one matrix” claim is *refuted* (a live depth- L stack has degree 3^L ; softmax is not a $\odot$) (proved). O’Neill’s stacked-linear-attention free monad is the free $\triangleleft$ -monoid on a linear container = the tensor algebra, correcting O’Neill Thm 3.2 (an

Adámek chain of partial sums, not bare powers) (proved). The `Mat(Vec)/attention` lane is the nearest proved anchor for GA composition; a dedicated GA/Kendall’s-*W* registry does not yet exist.

- **Game comonads are polynomial comonads.** Ehrenfeucht–Fraïssé, pebbling, and modal comonads factor through `Poly`; the underlying functor is polynomial (proved) and the comonoid is a small category (proved, with a 0-axiom EF-comonoid Lean witness). The arboreal-fibration = `Fam(codop)` crown remains speculative; the root is honestly computed.

Honest scope note: blockchain is covered via the orchestration re-entrancy result; tax computation and a standalone GA composition have *no* dedicated registry yet. Clio’s crystal-base results are off-territory and recorded only at her peer-claimed evidence level.

5 Formalisation

The registry contains roughly **107 lean-verified nodes**: sorry-free Lean 4 declarations whose axiom footprints are mostly `[propext]` or `[propext, Quot.sound]`. Four registries are lean-verified at the *root*: the free $\triangleleft$ -monoid three laws (`Container.freeMonoid`; `graft_assoc`, `split_assoc` use `[Quot.sound]` only), and the three H^2 engines $H^2(C_n; \mathbb{Z}) = \mathbb{Z}/n$, $H^2(C_n; \mathbb{Z}/m) = \mathbb{Z}/\gcd$, $H^2(\mathbb{Z}/2; \mathbb{Z}/2) = \mathbb{Z}/2$.

Notable lean nodes under other roots:

- **Equivalence chain:** M1–M4, ZS associativity, and all four monoidal structures, machine-checked in the core typeclass `Lean` (§2).
- **Tangent arc:** the oracles are `[propext]`-only—solid-idempotent collapse, universality rank-rigidity, two-model distinction—with the negation-obstruction arithmetic core at `[propext, Quot.sound]` (via `omega`) and a lean-verified positive companion for the $\mathbb{Z}$ far-side. *Caveat (registry README)*: Lean verifies the statement *as written*; e.g. the $\mathbb{N}$ -transport/functoriality in the scalar-object result is *not* formalised—the Lean node is the arithmetic core only.
- **ZS/ H^2 :** re-entrancy $[\omega] = \varepsilon$ (`Reentrancy.phi_omega`), o_ν at $|G| = 8$ and the $|G| = 16$ lock, the torsor corrected/converse (`Mathlib`), prunability descent L2.
- **Effect-coeffect:** `biKleisli` unit laws, `Maybe/Writer/affine arr_assoc`, the branching obstruction (negative witness), `Reader/State` drop, the sigma-monad, the proof-relevance boundary, A-module Fubini.
- **Linear containers:** the `Vec` collapse iso, `linExt_not_faithful`, the union-of-subspaces F_q core.
- **Game comonads:** the EF-comonoid witness (`#print axioms empty`), linear order $k = 2, 3, 4$.
- **Misc:** holonomy-triviality (full), the disjointness lemma, free/cofree UP, the `Workers` retract, state ΔS .

Treat lean-verified as “strongest check on what is on the page,” not “nothing left to review.”

6 Status & honesty

Top headline results with honest registry grades, and the honesty flags.

Result	Grade
Equivalence-chain core M1–M4 (full chain root, incl. M5 enriched / M6 ∞)	lean-verified <i>in-progress</i>
T1: $\llbracket - \rrbracket$ full & faithful $\Leftrightarrow$ unit connected	proved
∂ is the differential combinator of a CDC / tangent str.	proved (0-axiom finite bridge)
Tangent uniqueness: exactly two representable structures	proved
No scalar ring object over $\mathbb{N}$ (Cartan does not descend)	proved [propext, Quot.sound]
$\mathbb{Z}$ far side: $\mathbf{SPoly}_{\mathbb{Z}}$ Rosický, full Cartan on	proved (lean positive companion)
Lanfranchi parallelizable port ill-posed	proved (impossibility)
Pairwise ZS criterion $(L) \wedge (G)$	proved
o_ν homomorphism, $\text{im } \varphi^+ = \ker o_\nu$	proved (lean $ G = 8, 16$)
Holonomy-composition ZS bridge	peer-reviewed
Change-of-base $\mathcal{CL} \cap \mathcal{UN} = \{\text{Id}\}$	peer-reviewed
Orchestration / re-entrancy $[\omega] = \varepsilon$ in $H^2 \cong \mathbb{Z}/2$	lean-verified
Supply chain directed container + inventory $[\omega]$	computed
Game comonads are polynomial comonads	computed (root); proved + 0-axiom witness (parts)

Honesty flags.

1. *No self-published work.* The registry’s 18 published nodes are external literature cited as premises (Niu–Spivak, Schreier, Stallings–Swan, Ahman–Bauer, Baues–Wirsching, . . .), not MacBeth papers. Our shipped write-ups are registered at most as **peer-reviewed** (by Rick).
2. *Full chain in-progress.* M1–M4 lean-verified; M5 enriched unclassified; M6 ∞ speculative. Say “1-categorical core,” never “chain proved.”
3. *Tangent-uniqueness corrected.* Per Rick’s accepted referee report the ambient operation is Weil base change $(-) \otimes W$, *not* $(-) \triangleleft \mathbb{D}$ on full $(\text{Poly}, \triangleleft)$; the $\triangleleft$ -reading was wrong and F4 is demoted to open.
4. *Node vs theorem counts.* “**proved: 518**” is a *node* count (premises + sub-lemmas). Distinct top-level theorems number ~ 45 (one per registry root), of which 33 roots are **proved** or stronger and closed.
5. *Standing failure mode.* Write-ups have historically upgraded hedged proofs into clean false claims; grades here deliberately track registry nodes, not prose.

7 Open questions & roadmap

- **Full equivalence chain.** Close M5 (enriched $\mathbf{DCont} = \mathbf{Cof}$ —currently unclassified) and M6 (the ∞ -level $\infty\text{-DCont} = \mathbf{Cat}$, Kan–Chen Conj. 7.2—speculative). This is the single highest-value Theory-pillar target: it turns “core lean-verified” into “chain proved.”

- **Lie-group-object sub-question for Poly comonoids.** The Lanfranchi port is ill-posed as stated; the open question is whether a composition-*sensitive* invariant exists that recovers a groupoid criterion once T is honestly $\otimes W$ rather than $\triangleleft \mathbb{D}$.
- **The internal replacement theorem.** A named grant deliverable; not yet a registry root.
- **Path-5 applications.** Promote supply chains from `computed` to a worked real directed container; build the missing GA-composition (Kendall's W) and tax-computation registries rather than leaning on the attention lane as a proxy.
- **Strathclyde mini-course (Oct 13–16).** Neil teaching; the tangent/Cartan arc is the natural Theory-pillar lead—existence, the two-structure uniqueness, and the $\mathbb{N}/\mathbb{Z}$ Cartan knife are a clean, mostly machine-checked narrative that also surfaces the honest open frontier (the far-side Cartan descent and the Lie-object question).